

Assignment 5A: Airline Delays

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2026-09-30

Task

Submitted data file with at least some count analysis (+45)

Recreated File in Same Format as given, including missing data where there were empty cells in source data. (+5)

Placed file in Internet-accessible location, such as a publicly-accessible GitHub repo. Also OK (if less scalable) to generated and populated data frame from code. (+5)

Provided code to populate missing data. (+5)

Transformed data from wide to long format. (+10)

Compared percentage (not just counts) of either delays or arrival rates for two airlines overall. [Either with a chart, or a table; both would be exemplary]. Include text summarizes findings from this comparison. (+10)

Compared percentage (not just counts) of either delays or arrival rates for two airlines across five cities. [Either with a chart, or a table; both would be exemplary]. Include text summarizes findings from this comparison. (+10)

Describe discrepancy between comparing two airlines' flight performances city-by-city and overall. (+5)

Assignment Tidying and Transforming Data

Explain discrepancy between comparing two airlines' flight performances city-by-city and overall. (+5)

The chart above describes arrival delays for two airlines across five destinations. Your task is to: (1) Create a .CSV file (or optionally, a MySQL database!) that includes all of the information above. You're encouraged to use a "wide" structure similar to how the information appears above, so that you can practice tidying and transformations as described below. (2) Read the information from your .CSV file into R, and use tidy and dplyr as needed to tidy and transform your data. (3) Perform analysis to compare the arrival delays for the two airlines.

Approach

Step 1: Create database and table to insert given two airlines delay data (wide structure data) in PostgreSQL.
Step 2: Connect R to PostgreSQL Step 3: Read a SQL table into R Step 4: Save the SQL table as CSV
Step 5: Read wide structure data from github into R. Step 6: Transform wide data to long structure data.
Step 7: Compare two airlines arrival delays across five cities using plot percentage

Anticipated Challenges

converting wide structure data to long structure. Making comparison between two airlines across all cities.

Code Base

Load necessary libraries

```
library(DBI)
library(RPostgres)
library(tidyverse)

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.0      v readr      2.2.0
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2    4.0.3      v tibble    3.3.1
## v lubridate  1.9.5      v tidyr     1.3.2
## v purrr      1.2.1
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

Create Database, Table, and Insert Data

The following SQL code creates the `airlines_arrival_delay` table, inserts the data provided in the assignment, and displays all the records in the table.

```
CREATE TABLE airlines_arrival_delay("airline" VARCHAR(50), status VARCHAR(50), los_angeles
Int,phoenix Int, san_diego Int,san_francisco Int,seattle Int); INSERT INTO airlines_arrival_delay(airline,status,los_angeles
('ALASKA','on time',497,221,212,503,1841), (NULL,'delayed',62,12,20,102,305), (NULL, NULL, NULL,
NULL, NULL, NULL, NULL), ('AMWEST','on time',694,4840,383,320,201), (NULL,'delayed',117,415,65,129,61);
SELECT * FROM airlines_arrival_delay;
```

Connect R to PostgreSQL

```
con <- dbConnect(
  RPostgres::Postgres(),
  dbname = "Airlines_delay",
  host = "localhost",
  port = 5432,
  user = "postgres",
  password = Sys.getenv("DB_PASSWORD")
)
```

```
dbListTables(con)
```

```
## [1] "airlines_arrival_delay"
```

```
airlines_df <- dbReadTable(con, "airlines_arrival_delay")
head(airlines_df)
```

```
##   airline  status los_angeles phoenix san_diego san_francisco seattle
## 1  ALASKA on time          497      221        212           503      1841
## 2   <NA> delayed           62       12         20           102       305
## 3   <NA>   <NA>           NA       NA         NA            NA        NA
## 4  AMWEST on time          694     4840        383           320       201
## 5   <NA> delayed          117      415         65           129        61
```

```
#set first and second columns name empty.
names(airlines_df)[1:2] <- c(" ", " ")
write.csv(airlines_df, file="airlines_arrival_delay.csv", row.names = FALSE)
```

Load airline arrival delay dataset

```
url<-"https://raw.githubusercontent.com/lhamo07/Data-607-Assignment/refs/heads/main/Assignment-5A/airlines_arrival_delay.csv"
airlines_arrival_delay_df<-read.csv(file=url)
airlines_arrival_delay_df
```

```
##      X.      X..1 los_angeles phoenix san_diego san_francisco seattle
## 1 ALASKA on time          497      221        212           503      1841
## 2  <NA> delayed           62       12         20           102       305
## 3  <NA>   <NA>           NA       NA         NA            NA        NA
## 4  AMWEST on time          694     4840        383           320       201
## 5  <NA> delayed          117      415         65           129        61
```

Rename the first two un-labelled columns as “airline” and “status”

```
colnames(airlines_arrival_delay_df)[1:2] <- c("airline", "status")
airlines_arrival_delay_df
```

```
##   airline  status los_angeles phoenix san_diego san_francisco seattle
## 1  ALASKA on time          497      221        212           503      1841
## 2   <NA> delayed           62       12         20           102       305
## 3   <NA>   <NA>           NA       NA         NA            NA        NA
## 4  AMWEST on time          694     4840        383           320       201
## 5   <NA> delayed          117      415         65           129        61
```

```
airline_long_structure<-airlines_arrival_delay_df%>%
  fill(airline)%>%
  pivot_longer(cols=!(airline:status),
               names_to="city",
               values_to="flight_count",
               values_drop_na = TRUE
  )
airline_long_structure
```

```
## # A tibble: 20 x 4
##   airline status  city      flight_count
##   <chr>   <chr>  <chr>          <int>
```

```
## 1 ALASKA on time los_angeles 497
## 2 ALASKA on time phoenix 221
## 3 ALASKA on time san_diego 212
## 4 ALASKA on time san_francisco 503
## 5 ALASKA on time seattle 1841
## 6 ALASKA delayed los_angeles 62
## 7 ALASKA delayed phoenix 12
## 8 ALASKA delayed san_diego 20
## 9 ALASKA delayed san_francisco 102
## 10 ALASKA delayed seattle 305
## 11 AMWEST on time los_angeles 694
## 12 AMWEST on time phoenix 4840
## 13 AMWEST on time san_diego 383
## 14 AMWEST on time san_francisco 320
## 15 AMWEST on time seattle 201
## 16 AMWEST delayed los_angeles 117
## 17 AMWEST delayed phoenix 415
## 18 AMWEST delayed san_diego 65
## 19 AMWEST delayed san_francisco 129
## 20 AMWEST delayed seattle 61
```

First, I used `fill(airline)` to fill in the missing airline names with the corresponding airline name above. Then, I used `pivot_longer()` to convert the data from a wide structure to a long structure. In `cols = !(airline:status)`, all columns except the columns from airline through status are selected for transformation. The selected column names are stored in the new city column using `names_to = "city"`, and their corresponding values are stored in the flight_count column using `values_to = "flight_count"`. Finally, `values_drop_na = TRUE` removes rows where the flight count value is missing NA `### Percentage of delays rates for ALASKA airlines`

```
alaska_summary<-airline_long_structure%>%
filter(airline=="ALASKA")%>%
  summarize(
    airline="ALASKA",
    total_delayed=sum(flight_count[status=="delayed"]),
    total_on_time = sum(flight_count[status == "on time"]),
    grand_total = total_delayed + total_on_time,
    delay_percent = round((total_delayed / grand_total) * 100,digits=2)
  )
alaska_summary
```

```
## # A tibble: 1 x 5
##   airline total_delayed total_on_time grand_total delay_percent
##   <chr>         <int>         <int>         <int>         <dbl>
## 1 ALASKA           501           3274           3775           13.3
```

Percentage of delays rates for AMWEST airlines

```
amwest_summary<-airline_long_structure%>%
filter(airline=="AMWEST")%>%
  summarize(
    airline="AMWEST",
```

```

total_delayed=sum(flight_count[status=='delayed']),
total_on_time = sum(flight_count[status == "on time"]),
grand_total   = total_delayed + total_on_time,
delay_percent = round((total_delayed / grand_total) * 100,digits=2)
)
amwest_summary

```

```

## # A tibble: 1 x 5
##   airline total_delayed total_on_time grand_total delay_percent
##   <chr>         <int>         <int>         <int>         <dbl>
## 1 AMWEST             787             6438             7225             10.9

```

```

percentage_summary<-bind_rows(alaska_summary,amwest_summary)
percentage_summary

```

```

## # A tibble: 2 x 5
##   airline total_delayed total_on_time grand_total delay_percent
##   <chr>         <int>         <int>         <int>         <dbl>
## 1 ALASKA             501             3274             3775             13.3
## 2 AMWEST             787             6438             7225             10.9

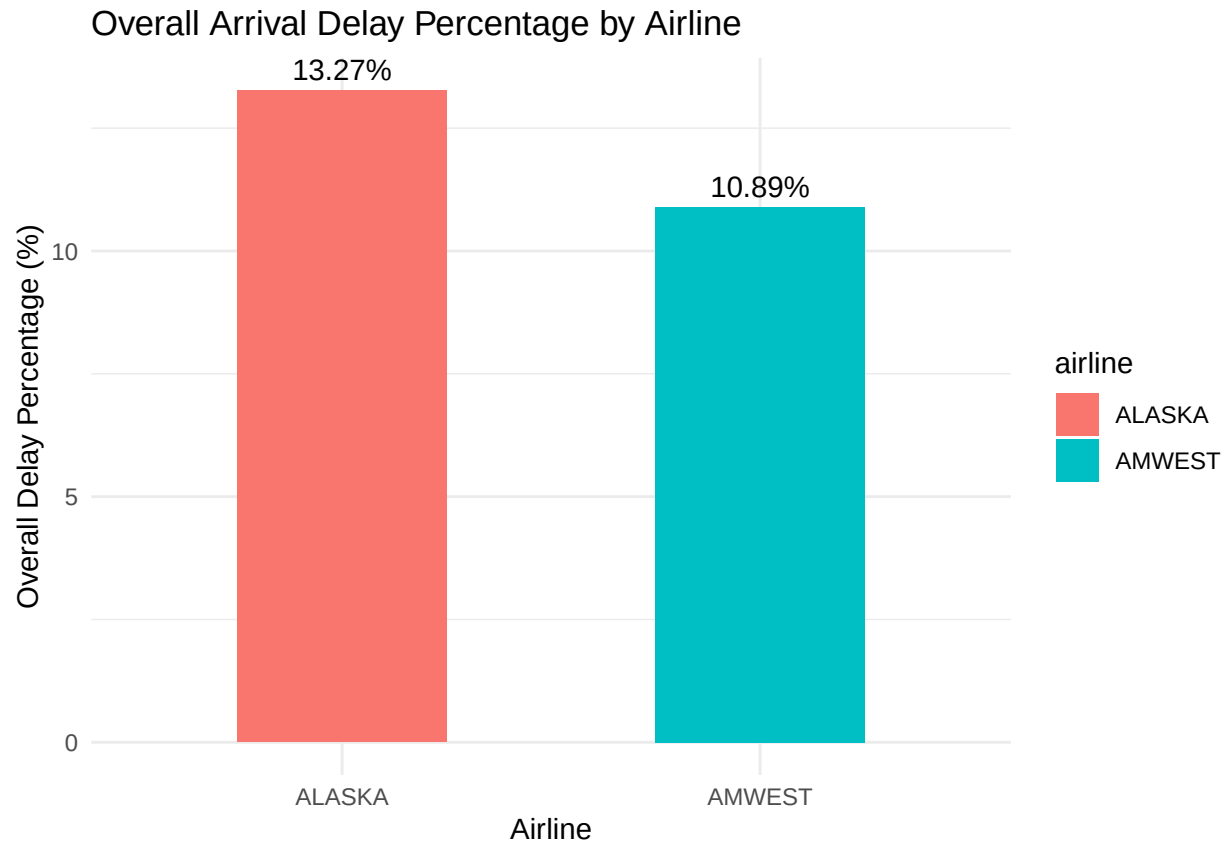
```

Compared percentage of delays rates for two airlines overall.

```

ggplot(percent_summary,aes(x=airline,y=delay_percent,fill=airline))+
  geom_col(width=0.5)+
  geom_text(
    aes(label = paste0(delay_percent, "%")),
    vjust = -0.5)+
  labs(
    title = "Overall Arrival Delay Percentage by Airline",
    x = "Airline",
    y = "Overall Delay Percentage (%)"
  )+
  theme_minimal()

```



From the above plot, I found that the overall delay percentage for ALASKA Airline is 13.27%, while AMWEST Airline is 10.83%. Based on this comparison, I would say that AMWEST Airlines appears to be more reliable for passengers who are concerned about flight delays.

Compared percentage of delays rates for two airlines across five cities.

```
city_summary<-airline_long_structure%>%
group_by(airline,city)%>%
  mutate(city_total=sum(flight_count),
         delay_percentage=round((flight_count/city_total)*100,digits=2))%>%

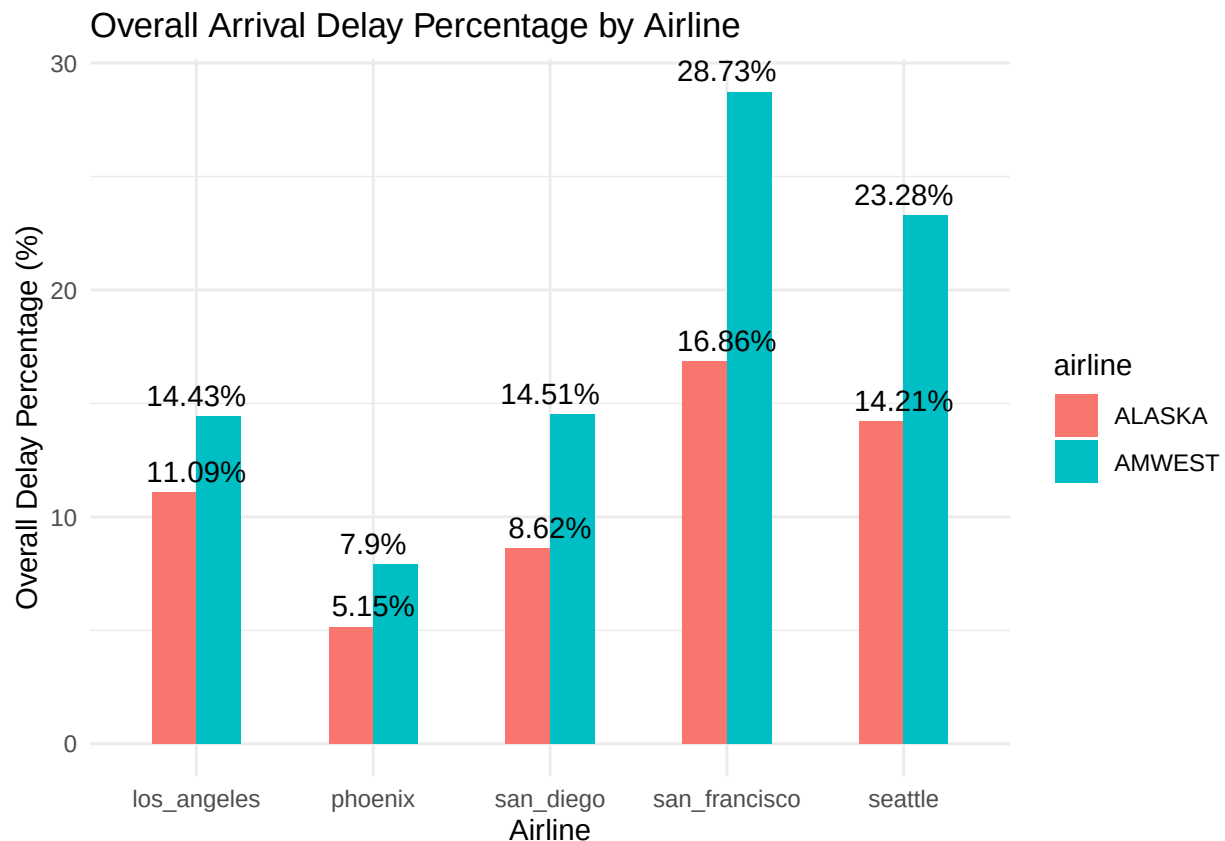
  filter(status=="delayed")

city_summary
```

```
## # A tibble: 10 x 6
## # Groups:   airline, city [10]
##   airline status  city      flight_count city_total delay_percentage
##   <chr>    <chr>  <chr>         <int>      <int>         <dbl>
## 1 ALASKA  delayed los_angeles      62        559           11.1
## 2 ALASKA  delayed phoenix         12        233            5.15
## 3 ALASKA  delayed san_diego      20        232            8.62
```

```
## 4 ALASKA delayed san_francisco 102 605 16.9
## 5 ALASKA delayed seattle 305 2146 14.2
## 6 AMWEST delayed los_angeles 117 811 14.4
## 7 AMWEST delayed phoenix 415 5255 7.9
## 8 AMWEST delayed san_diego 65 448 14.5
## 9 AMWEST delayed san_francisco 129 449 28.7
## 10 AMWEST delayed seattle 61 262 23.3
```

```
ggplot(city_summary,aes(x=city,y=delay_percentage,fill=airline))+
  geom_col(width=0.5,position="dodge")+
  geom_text(
    aes(label = paste0(delay_percentage, "%")),
    vjust = -0.5)+
  labs(
    title = "Overall Arrival Delay Percentage by Airline",
    x = "Airline",
    y = "Overall Delay Percentage (%)"
  )+
  theme_minimal()
```



From the above plot, I can conclude that Phoenix has the lowest flight delay percentage for both airlines. San Francisco has the highest delay percentage for AMWEST Airlines. The remaining cities have relatively similar flight delay percentages for both airlines.

Conclusion

Although the overall percentage of delays for ALASKA Airlines is higher than AMWEST Airlines, the city level comparison shows that AMWEST has a higher delay percentage in some cities. Therefore, if I were planning a trip, I would compare the delay percentages by city rather than relying only on the overall delay percentage.

citation

<https://chatgpt.com/share/6abc7bac-d99c-83e9-a515-e584ba7ef430> <https://chatgpt.com/share/6ac2d87e-7cb8-83ea-899e-9e8371c0fbb0> OpenAI. (2026). ChatGPT [Large language model]. <https://chatgpt.com>. Accessed October 4, 2026.